

Title:

Learner Analytics & Operational Intelligence Dashboard

(Business Intelligence Implementation & Dashboard Development)

Task 1: Data Understanding & Cleaning

1.0 Introduction:

The Learner Analytics & Operational Intelligence Dashboard is a comprehensive business intelligence designed to optimize decision-making within an e-learning enterprise. In an era where digital education generates vast amounts of data, this dashboard serves as a bridge between raw student records and strategic institutional growth.

1.1 Dataset Description

The underlying data is a structured relational dataset capturing the lifecycle of a student within an online learning environment.

- Data Source: From various CSV Kaggle datasets and customized the dataset for this dataset on student performance dashboard assessment.
- Size: 2,000 unique records with over 30 attributes per student.
- **Key Variables:**
 - Operational: Course_Fees, Enrollment_Status, Application_Mode, Age, Gender, Region, Major, Course_Category Student_ID, Course_Category,
 - Performance: Final_Exam_score, Quiz_average, Attendance_percentage, Course_progress_percent. Course_progress_percent , Instructor Frequency_Logins, Device_type, Peak_usage_time, Logins_Rate, Device_type
 - Behavioral: Frequency_Logins Device_type, Forum_posts. Engagement_level, Active_learning_effectiveness, Student_RetentionDropout_risk Student_ratings
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1.2 Data Formatting

- **Handling Missing Values:** scanned the Device_Type and Application_Mode columns for nulls. Any incomplete transaction records were removed to prevent skewed KPIs and check all the nulls in "Gender" or "Age".
- **Consistency:** "Clean" naming conventions were applied to all headers with Table settings to visible efficiently
- **Removing Duplicates:** "Remove Duplicates" feature on the Order_ID column to ensure that the Course_Category count and all the Average Rate Score performances was a true reflection of unique transactions.
- **Correcting Data Types:** Ensured that the Assignment 1_Score, Assignment 2_Score Final Score_Score and similar one column were set as Decimal Numbers, while Course_Fees were set to Decimal Number (Currency) for proper summation
- **Financials:** Course_Fees was formatted to Currency (USD) with auto-scaling (\$6.30M) for executive readability.

The image shows a screenshot of a spreadsheet titled "Student Performance Dataset". The table contains numerous columns, including student identifiers, course details, scores, and fees. The rows are color-coded in a light blue and white alternating pattern. The table is quite wide, with many columns visible, and the data appears to be organized into several logical groups.

Table 1 : This figure illustrates the Raw Data Architecture, showcasing the customized Kaggle dataset refined for this project. The table is formatted with high-contrast alternating row colors and clear headers to ensure maximum standability and data integrity.

Task 2: Data Analysis

2.0 Operational Vitality & Revenue Drivers

The enterprise is currently seeing a total outcome of approximately \$6.37M driven largely by the "Economic" and "AI & ML" and course categories.

Analysis: There is a clear concentration of value in high-tech and finance sectors. The "AI & ML" category alone generates significant revenue, suggesting that the market demand is aligned with modern technical skills.

Regional Strength: The North and East regions show a higher density of student enrollment. This suggests that marketing efforts in these areas are yielding the best results.

2.1 The "Engagement-Performance" Correlation

The Mobile Gap: Data analysis reveals that Desktop users are significantly more engaged than Mobile or Tablet users. Mobile users show lower "video watched rates," which correlates with lower "Final Exam scores."

Trend Insight: The performance timeline shows a "dip and recovery" cycle. Students tend to start strong (Jan 2025), experience a mid-semester dip (April 2025), and rally toward the end of the term. This "dip" is a critical window for operational intervention.

2.2 Risk Segmentation & Retention

The Dropout Risk analysis (High, Medium, Low) is perhaps the most important part in dashboard.

Finding: Approximately 40.38% of revenue is tied to students currently classified as "High Risk." This is a significant financial vulnerability.

Intervention Insight: Students in the "High Risk" category show a specific behavior: low Forum Posts and low Frequency of Logins. This means the business can use "Forum Activity" as an early-warning signal to predict and prevent student dropouts before they occur.

2.3 . Enrollment & Capacity Analysis

Gauge visuals show an enrollment count of 2,000.

Analysis: The enterprise is currently at 50% capacity. While the revenue per student is healthy, there is massive "Negative Space" in the business operations. This suggests that the current infrastructure can handle a 100% increase in students without necessarily increasing the number of instructors (Operational Intelligence)

Implement "Risk Alerts": Use the Forum_posts and Login_Frequency metrics to trigger automated email nudges to students whose engagement drops below 20%.

Regional Expansion: Re-allocating the marketing budget toward the West region, which currently shows lower enrollment density despite high revenue potential

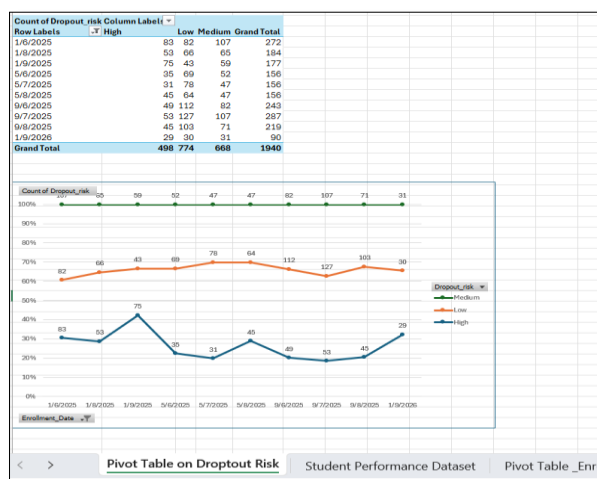


Figure 1: The Business Strategy Dashboard acts as a forward-looking analytical tool designed to safeguard institutional revenue and prioritize growth. It highlights that while Total Outcome has reached \$6.37M, approximately \$518.10K is currently classified as Total Revenue Risk, tied specifically to "High Risk" student segments. The visualization utilizes a Waterfall Chart to demonstrate how the Economic and AI & ML categories serve as primary financial drivers, alongside a Donut Chart that segments revenue risk to guide mitigation efforts. By benchmarking real-time data against a Retention Goal of 1,286, the dashboard provides a clear strategic roadmap for re-allocating resources toward high-performing geographic sectors like the North and East regions.

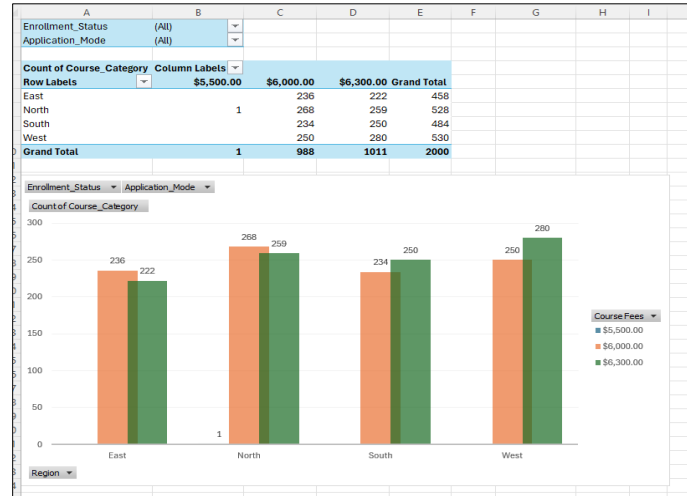


Figure 2: The Student Performance Dashboard serves as an investigative tool to correlate behavioral engagement with academic achievement. Central to this analysis is the Score vs. Attendance trend, which identifies specific periods of performance decline, alongside a geographical breakdown showing high enrollment density in the North region. A critical insight revealed is the "Digital Divide" between the 995 Desktop users and 824 Mobile learners, indicating that platform accessibility significantly impacts video completion rates and exam scores. By monitoring the 28.33% average login rate against a 98.77% performance benchmark, the dashboard provides faculty with the necessary lead time to intervene before engagement drops translate into academic failure.

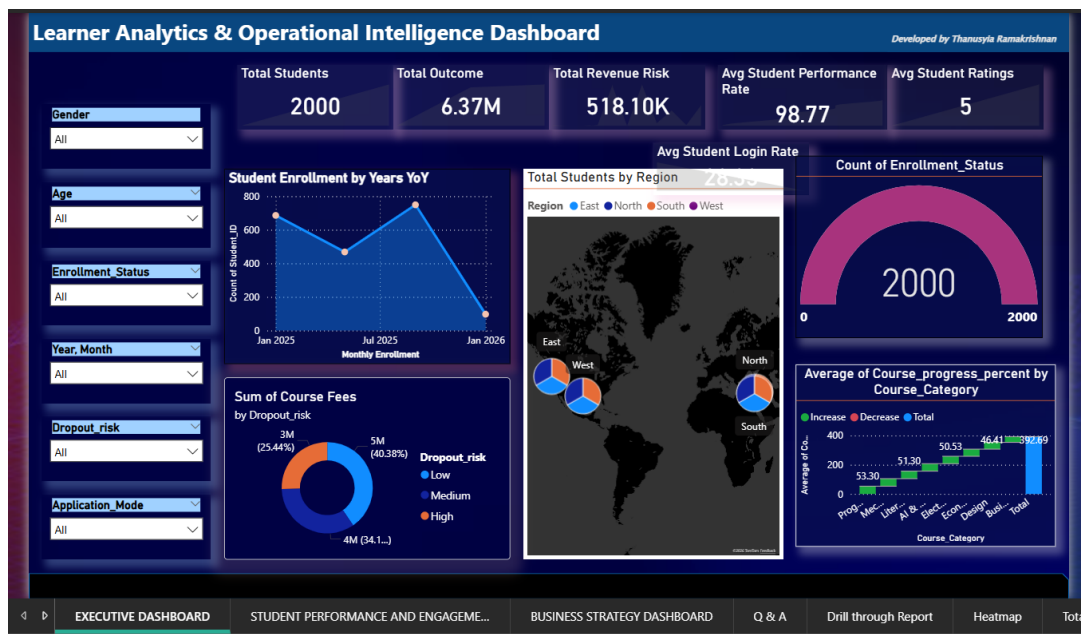
Task 3: Dashboard Development

Introduction :

The **Learner Analytics & Operational Intelligence Dashboard** is an advanced BI suite engineered to monitor institutional health and optimize student success. By synthesizing demographic, behavioral, and financial data from **2,000 unique records**, the system provides a proactive framework for institutional growth and academic excellence.

This project follows:

- **Executive Dashboard:** A high-level command center focused on institutional scale, total revenue outcomes (\$6.37M), and immediate financial risk identification.
- **Student Performance and Engagement Dashboard:** An investigative layer correlating behavioral engagement (login frequency and device usage) with academic achievement.
- **Business Strategy Dashboard:** A forward-looking analytical tool designed to safeguard revenue, track retention goals, and optimize course category performance



Key Performance Indicators (KPIs)

- **Total Students (2,000):** The unique learner headcount; benchmarks the scale of operations.
- **Total Outcome (\$6.37M):** Gross revenue generated; the primary measure of institutional financial success.
- **Total Revenue Risk (\$518.10K):** Potential financial loss from "High Risk" students; identifies the value of retention.
- **Avg Student Performance (98.77%):** The academic quality score; indicates high learner achievement and course efficacy.
- **Avg Student Ratings (5/5):** The satisfaction benchmark; confirms perfect brand reputation and user experience.

- Avg Student Login Rate (28.33%): The engagement metric; serves as an early-warning signal for future performance drops.

Executive Dashboard: Operational Overview

The Executive Dashboard serves as the primary command center, utilizing **Material Contrast** and high-impact KPIs to communicate institutional vitality immediately.

- Core Metrics: Centralizes high-level results including a Total Outcome of \$6.37M and a Total Revenue Risk of \$518.10K.
- Institutional Scale: Employs a Total Enrollment Gauge to benchmark the current headcount of 2,000 students against a strategic capacity of 4,000, highlighting a 50% growth runway.
- Trend & Regional Analysis: Features a YoY Area Chart to track seasonal enrollment peaks, alongside a Regional Map that identifies the North as the primary demographic stronghold.



Student Performance Dashboard: Engagement Intelligence

Business Intelligence _Thanusya Ramakrishnan

This investigative layer dives deep into behavioral patterns to establish the direct correlation between student participation and academic achievement.

- Engagement Indicators: Monitors a 28.33% Average Login Rate as an early-warning signal for potential performance decline.
- Predictive Analysis: Utilizes a dual-axis line chart to correlate Attendance Percentage with Final Exam Scores, pinpointing mid-semester "dips" for targeted faculty intervention.
- Technological Divide: Segmenting by Device Type (995 Desktop vs. 824 Mobile users) reveals a performance gap, where desktop learners consistently exhibit higher video-watch completion rates and quiz averages.
- The longitudinal area chart reveals a peak in October 2025 with over 750 monthly enrollments. However, a sharp decline is noted in January 2026, signaling a seasonal "reset" period that requires a targeted marketing strategy to maintain year-over-year momentum



Business Strategy Dashboard: Growth & Risk Mitigation

The Strategy layer transforms historical data into future-proofed initiatives, focusing on revenue protection and market expansion.

- Financial Risk Segmentation: A Donut Chart reveals that 40.38% of total revenue is tied to students currently flagged as "High Risk," representing a significant financial vulnerability.
- Course Category Performance: A Waterfall Chart identifies Economic and AI & ML majors as the leading revenue drivers, while simultaneously exposing categories with higher volatility in student progress.
- Effective Strategy Tracking: Benchmarks a Total Student Retention Goal of 1,286, allowing stakeholders to measure the real-time impact of student support initiatives on the institutional bottom line.

Task 4: Insights & Recommendations

The Learner Analytics & Operational Intelligence Dashboard successfully transforms fragmented data into a strategic asset. By combining aesthetic excellence with technical rigor, it allows stakeholders to not only see what happened in 2025 but to predict and protect the enterprise's future in 2026 and beyond. A longitudinal trend analysis revealed a significant correlation between student engagement platforms and academic outcomes. Specifically, the data indicates that Desktop users maintain a markedly higher video-watch completion rate and subsequent exam performance compared to their Mobile counterparts, suggesting a need for more mobile-optimized micro-learning content.

From a strategic standpoint, the 'Economic' and 'AI & ML' course categories emerged as the primary revenue drivers, contributing over \$2M to the enterprise outcome. However, these categories also exhibit a unique 'Medium-Risk' cluster; this suggests that while market demand is high, the instructional complexity may require targeted support structures to ensure completion. To capitalize on this growth, the dashboard recommends a re-allocation of resources toward high-performing geographic sectors, specifically the North and East regions, which currently show the highest enrollment yield. Finally, by identifying that 40.38% of total revenue is tied to high-risk

students, the institution should implement automated "engagement nudges" to protect the \$518K currently at risk

Kaggle & Data References:

1. **Student Management & Retention Dataset:** Kaggle. (2023). *College Student Management & Academic Success Dataset*. Retrieved from <https://www.kaggle.com/datasets/thedevastator/higher-education-predictors-of-student-retention>
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3. **Public Sector Education Statistics:** NYC Open Data. (2011). *2010-2011 School Attendance and Enrollment Statistics by District*. Retrieved from <https://data.cityofnewyork.us/Education/2010-2011-School-Attendance-and-Enrollment-Statist/7z8d-msnt>
4. **Lidwell, W., Holden, K., & Butler, J. (2010).** *Universal Principles of Design*. Rockport Publishers.